Evaluation runs: **bank** (bank-grove)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

No run of the stateless engine on this benchmark.

stateful

No run of the stateful engine on this benchmark.

offline

offline: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	366	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	30	43383	conforms	2.0	22	10 min, 16 GB	0.0	2.0	0.0	in-memory	brain05, 2026-11-02
3	140	2752874	conforms	117.9	65	10 min, 16 GB	0.0	117.8	0.1	in-memory	brain05, 2026-11-02

offline: pair **unordered**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	258	9	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	30	11072	157	violation	0.6	18	10 min, 16 GB	0.0	0.6	0.0	in-memory	brain05, 2026-11-02
3	140	247764	1586	violation	7.8	44	10 min, 16 GB	0.0	7.7	0.1	in-memory	brain05, 2026-11-02
4	630	4869014	14510	violation	227.0	238	10 min, 16 GB	0.1	225.6	1.4	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 unordered fifo.

skipped: a smaller size had reached its limit: 4 ordered fifo, 5 ordered fifo, 5 unordered fifo, 6 unordered fifo, 6 ordered fifo, 7 unordered fifo, 7 ordered fifo, 8 ordered fifo, 8 unordered fifo, 9 ordered fifo, 9 unordered fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	366	conforms	8.3	211	48 h, 80 GB	0.1	8.3	0.0	merge	pinky12, 2026-11-07
2	30	43383	conforms	141.7	11647	48 h, 80 GB	0.2	141.7	0.0	merge	pinky07, 2026-11-07

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	140	2752874	conforms	20843.2	80190	48 h, 250 GB	0.4	20843.0	0.1	merge	brain01, 2026-11-07

cross offline, Java implementation against the Rust specification: pair **unordered**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	258	9	violation	7.5	221	48 h, 80 GB	0.1	7.5	0.0	merge	pinky12, 2026-11-07
2	30	11072	157	violation	49.3	7538	48 h, 80 GB	0.2	49.3	0.0	merge	pinky07, 2026-11-07
3	140	247764	1586	violation	1972.0	19800	48 h, 250 GB	0.4	1971.8	0.1	merge	brain01, 2026-11-07

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	366	conforms	3.1	169	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	30	43383	conforms	3.1	178	10 min, 16 GB	3.1	2.0	0.0	in-memory	brain01, 2026-11-02
3	140	2752874	conforms	117.8	217	10 min, 16 GB	3.6	117.8	0.1	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **unordered**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	258	9	violation	3.1	169	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	30	11072	157	violation	3.1	174	10 min, 16 GB	3.1	0.6	0.0	in-memory	brain01, 2026-11-02
3	140	247764	1586	violation	7.8	196	10 min, 16 GB	3.6	7.7	0.1	in-memory	brain01, 2026-11-02

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 4 fifo unordered, 4 fifo ordered.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	6	366	conforms	10.3	342	1 h, 250 GB		brain05, 2026-11-02
2	30		(1 h)			1 h, 250 GB		brain05, 2026-11-02
3	140		(1 h)			1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **unordered**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	6	258	110	violation	5.2	322	1 h, 250 GB	349	brain05, 2026-11-02
2	30	11072	5598	violation	645.6	973	1 h, 250 GB	820	brain05, 2026-11-02
3	140			(1 h)			1 h, 250 GB		brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 1 fifo ordered, 1 fifo unordered, 2 fifo ordered, 2 fifo unordered, 3 fifo unordered, 3 fifo ordered.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	6	366	conforms	3.2	165	1 h, 250 GB		brain01, 2026-11-02
2	30	43383	conforms	5.1	160	1 h, 250 GB		brain01, 2026-11-02
3	140	2752874	conforms	112.7	156	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **unordered**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	6	258	110	violation	3.2	165	1 h, 250 GB	2	brain01, 2026-11-02
2	30	11072	5598	violation	3.6	160	1 h, 250 GB	2	brain01, 2026-11-02
3	140	247764	122182	violation	11.9	156	1 h, 250 GB	2	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	366	366	conforms	9.9	962	10 min, 16 GB	9.9	0.1	0.0	in-memory	brain01, 2026-11-02
2	43383	43383	conforms	1504.8	15259	10 min, 16 GB	1504.7	2.0	0.0	in-memory	brain01, 2026-11-02
3	2752874	2752874	conforms	20843.1	80244	48 h, 250 GB	20843.0	124.8	0.1	merge	brain01, 2026-11-04

twins offline, Rust implementation against its Java twin: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	258	258	conforms	6.4	333	10 min, 16 GB	6.4	0.1	0.0	in-memory	brain01, 2026-11-02
2	11072	11072	conforms	317.2	14584	10 min, 16 GB	317.1	0.6	0.0	in-memory	brain01, 2026-11-02
3	247764	247764	conforms	1972.0	19836	48 h, 250 GB	1971.8	9.6	0.2	merge	brain01, 2026-11-04

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 4 fifo unordered.

left out by the table rules (ok): 3 fifo unordered.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	366	366	conforms	9.9	962	10 min, 16 GB	0.1	9.9	0.0	in-memory	brain01, 2026-11-02
2	43383	43383	conforms	1504.8	15259	10 min, 16 GB	2.0	1504.7	0.0	in-memory	brain01, 2026-11-02
3	2752874	2752874	conforms	20843.1	80244	48 h, 250 GB	124.8	20843.0	0.1	merge	brain01, 2026-11-04

twins offline, Java implementation against its Rust twin: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	258	258	conforms	6.4	333	10 min, 16 GB	0.1	6.4	0.0	in-memory	brain01, 2026-11-02
2	11072	11072	conforms	317.2	14584	10 min, 16 GB	0.6	317.1	0.0	in-memory	brain01, 2026-11-02
3	2752874	247764	violation*	1972.0	19854	48 h, 250 GB	124.8	1971.8	0.1	merge	brain01, 2026-11-04

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 4 fifo unordered.
left out by the table rules (ok): 3 fifo unordered.

twins online, Rust implementation (TraceForge) against the Java twin's export

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3	2752874	2752874	conforms	20976.8	80188	48 h, 250 GB		brain01, 2026-11-04

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3	247764	247764	conforms	1982.0	19798	48 h, 250 GB		brain01, 2026-11-04

Not run (twins online, Rust implementation (TraceForge) against the Java twin's export). planned but never scheduled (grid points beyond the ladder): 1 fifo unordered, 1 fifo ordered, 2 fifo ordered, 2 fifo unordered.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	366	366	conforms	9.0	891	10 min, 16 GB	9.0	0.4	0.0	in-memory	brain02, 2026-11-03
2	43383	43383	conforms	1288.6	15339	10 min, 16 GB	1288.6	2.3	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	258	258	conforms	7.9	758	10 min, 16 GB	7.8	0.2	0.1	in-memory	brain02, 2026-11-03
2	11072	11072	conforms	240.5	14919	10 min, 16 GB	240.4	0.7	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). refused (an input missing at run time): 3 fifo unordered, 3 fifo ordered.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	366	366	conforms	9.0	891	10 min, 16 GB	0.4	9.0	0.0	in-memory	brain02, 2026-11-03
2	43383	43383	conforms	1288.6	15339	10 min, 16 GB	2.3	1288.6	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	258	258	conforms	7.9		10 min, 16 GB	0.2	7.8	0.1	in-memory	brain02, 2026-11-03
2	11072	11072	conforms	240.5	14919	10 min, 16 GB	0.7	240.4	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 3 fifo unordered, 3 fifo ordered.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	366	366	conforms	10.6	952	1 h, 250 GB		brain02, 2026-11-03
2	43383	43383	conforms	1308.1	15666	1 h, 250 GB		brain02, 2026-11-03
3	2752874		(1 h)			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	258	258	conforms	7.2	708	1 h, 250 GB		brain02, 2026-11-03
2	11072	11072	conforms	264.1	14822	1 h, 250 GB		brain02, 2026-11-03
3	247764		(1 h)			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	366	366	conforms	12.9	887	1 h, 250 GB		brain02, 2026-11-03
2	43383	43383	conforms	1290.8	15323	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	258	258	conforms	7.9	754	1 h, 250 GB		brain02, 2026-11-03
2	11072	11072	conforms	241.0	14903	1 h, 250 GB		brain02, 2026-11-03

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).
skipped: a smaller size had reached its limit: 3 fifo unordered, 3 fifo ordered.